Set 11 practice: Surface integrals and Stokes' theorem

Zill §9.13, 9.14

Quiz prep. Closed book, about 10 minutes per problem. Problems 1 to 4 are surface area and surface integrals of scalar functions (graphs first, then parametric surfaces), problems 5 and 6 are flux integrals, problem 7 verifies Stokes' theorem by computing both sides, and problem 8 uses Stokes' theorem to swap a curved surface for a flat one with the same boundary. Every answer was recomputed by a second route.

Key methods

Scalar surface integrals. For S the graph $z = g(x, y)$ over a region R of the xy-plane:

$$dS = \sqrt{1 + g_x^2 + g_y^2}\, dA, \qquad \iint_S f\, dS = \iint_R f(x, y, g(x, y))\sqrt{1 + g_x^2 + g_y^2}\, dA$$

For a parametric surface $\mathbf{r}(u, v)$ over a parameter region D:

$$dS = |\mathbf{r}_u \times \mathbf{r}_v|\, dA, \qquad \iint_S f\, dS = \iint_D f(\mathbf{r}(u, v))\, |\mathbf{r}_u \times \mathbf{r}_v|\, dA$$

Surface area is the case $f = 1$. Do not forget the factor $|\mathbf{r}_u \times \mathbf{r}_v|$ or the square root.

Flux. The flux of $\mathbf{F}$ through an oriented surface S with unit normal $\mathbf{n}$ is $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$. State which side $\mathbf{n}$ points to before computing.

- Graph $z = g(x, y)$, **upward** normal: $\mathbf{n}\, dS = \langle -g_x, -g_y, 1\rangle\, dA$. Downward is the negative.
- Parametric: $\mathbf{n}\, dS = \pm\, (\mathbf{r}_u \times \mathbf{r}_v)\, dA$, with the sign chosen so the normal points the way you were told. Check one point.

Stokes' theorem. Let S be a piecewise smooth oriented surface whose boundary C is a simple closed piecewise smooth curve, and let $\mathbf{F}$ have continuous first partials on an open region containing S. Then

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Orientation of C by the right-hand rule: with the thumb along $\mathbf{n}$, the fingers curl in the direction of travel on C. For an upward normal this means counterclockwise seen from above.

Each problem is on its own page with space for your working; the solution is on the next page.

Problem P1

Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4$.

Solution P1

Area of a graph

The surface is the graph $z = g(x, y) = x^2 + y^2$, with $g_x = 2x$ and $g_y = 2y$. The plane $z = 4$ meets it where $x^2 + y^2 = 4$, so the surface lies over the disk $R : x^2 + y^2 \le 4$.

$$dS = \sqrt{1 + 4x^2 + 4y^2}\, dA$$

Switch to polar coordinates, $dA = r\, dr\, d\theta$, $0 \le r \le 2$, $0 \le \theta \le 2\pi$:

$$A = \int_0^{2\pi} \int_0^2 \sqrt{1 + 4r^2}\, r\, dr\, d\theta$$

Let $u = 1 + 4r^2$, so $du = 8r\, dr$. The limits $r = 0, 2$ become $u = 1, 17$.

$$\int_0^2 \sqrt{1 + 4r^2}\, r\, dr = \frac{1}{8} \int_1^{17} u^{1/2}\, du = \frac{1}{8} \cdot \frac{2}{3}\left(17^{3/2} - 1\right) = \frac{17\sqrt{17} - 1}{12}$$

$$\boxed{A = 2\pi \cdot \frac{17\sqrt{17} - 1}{12} = \frac{\pi}{6}(17\sqrt{17} - 1) \approx 36.18}$$

Sanity check: the flat disk under the surface has area $4\pi \approx 12.57$, and the paraboloid is steep near the rim, so a surface area near three times that is reasonable.

Problem P2

Evaluate $\iint_S (x^2 + y^2)\, dS$, where S is the part of the cone $z = \sqrt{x^2 + y^2}$ with $1 \leq z \leq 2$.

Solution P2

Scalar integral over a graph

S is the graph of $g = \sqrt{x^2 + y^2}$ over the annulus $R : 1 \le x^2 + y^2 \le 4$. Away from the origin g is smooth, and the origin is not on S.

$$g_x = \frac{x}{\sqrt{x^2 + y^2}}, \quad g_y = \frac{y}{\sqrt{x^2 + y^2}}, \quad g_x^2 + g_y^2 = 1 \implies dS = \sqrt{2}\, dA$$

The integrand $f = x^2 + y^2$ does not involve z, so nothing needs substituting. In polar coordinates:

$$\iint_S (x^2 + y^2)\, dS = \sqrt{2} \int_0^{2\pi} \int_1^2 r^2 \cdot r\, dr\, d\theta = \sqrt{2} \cdot 2\pi \cdot \frac{2^4 - 1^4}{4} = \sqrt{2} \cdot 2\pi \cdot \frac{15}{4}$$

$$\boxed{\iint_S (x^2 + y^2)\, dS = \frac{15\sqrt{2}\,\pi}{2} \approx 33.32}$$

The $\sqrt{2}$ is the slope factor of a 45° cone: the cone is steeper than its shadow by exactly $1/\cos 45^\circ$.

Problem P3

The helicoid S is parametrized by $\mathbf{r}(u,v) = \langle u\cos v,\ u\sin v,\ v\rangle$ for $0 \le u \le 1$, $0 \le v \le \pi$. Find its surface area.

Solution P3

Area of a parametric surface

The surface winds upward like a spiral ramp, so it is not a graph over a single region of the xy-plane in any convenient way. Use the parametrization.

$$\mathbf{r}_u = \langle \cos v,\ \sin v,\ 0 \rangle, \qquad \mathbf{r}_v = \langle -u\sin v,\ u\cos v,\ 1 \rangle$$

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos v & \sin v & 0 \\ -u\sin v & u\cos v & 1 \end{vmatrix} = \langle \sin v,\ -\cos v,\ u\cos^2 v + u\sin^2 v \rangle = \langle \sin v,\ -\cos v,\ u \rangle$$

$$|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{\sin^2 v + \cos^2 v + u^2} = \sqrt{1+u^2}$$

The parameter region is the rectangle $0 \le u \le 1$, $0 \le v \le \pi$, so

$$A = \int_0^{\pi}\int_0^1 \sqrt{1+u^2}\, du\, dv = \pi \int_0^1 \sqrt{1+u^2}\, du$$

Use $\int \sqrt{1+u^2}\, du = \frac{1}{2}\Big[u\sqrt{1+u^2} + \ln\left(u + \sqrt{1+u^2}\right)\Big]$ (trigonometric substitution $u = \tan t$):

$$\int_0^1 \sqrt{1+u^2}\, du = \frac{1}{2}\Big[\sqrt{2} + \ln\left(1+\sqrt{2}\right)\Big]$$

$$\boxed{A = \frac{\pi}{2}\Big[\sqrt{2} + \ln\left(1+\sqrt{2}\right)\Big] \approx 3.608}$$

Sanity check: the shadow of the surface on the xy-plane is a half-disk of radius 1, area $\pi/2 \approx 1.571$. The ramp is tilted, so its area must be larger than that, and 3.608 is.

Problem P4

Evaluate $\iint_S x^2 z \, dS$, where S is the cylinder $x^2 + y^2 = 4$ with $0 \leq z \leq 3$ (the curved side only, no end caps).

Solution P4

Scalar integral where the surface is not a graph over the xy-plane

The cylinder is not a graph $z = g(x, y)$, since many points share one (x, y). Parametrize it by the angle and the height:

$$\mathbf{r}(\theta, z) = \langle 2\cos\theta,\ 2\sin\theta,\ z\rangle, \qquad 0 \le \theta \le 2\pi,\ \ 0 \le z \le 3$$

$$\mathbf{r}_\theta = \langle -2\sin\theta,\ 2\cos\theta,\ 0\rangle, \qquad \mathbf{r}_z = \langle 0,\ 0,\ 1\rangle$$

$$\mathbf{r}_\theta \times \mathbf{r}_z = \langle 2\cos\theta,\ 2\sin\theta,\ 0\rangle, \qquad |\mathbf{r}_\theta \times \mathbf{r}_z| = 2$$

So $dS = 2\,d\theta\,dz$, the circumference element $2\,d\theta$ times the height element. On S, $x^2 z = 4\cos^2\theta\ z$:

$$\iint_S x^2 z\,dS = \int_0^3\int_0^{2\pi} 4\cos^2\theta\ z\cdot 2\ d\theta\,dz = 8\Big(\int_0^{2\pi}\cos^2\theta\,d\theta\Big)\Big(\int_0^3 z\,dz\Big) = 8\cdot\pi\cdot\frac{9}{2}$$

$$\boxed{\iint_S x^2 z\,dS = 36\pi \approx 113.1}$$

Problem P5

Let S be the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy-plane, oriented **upward** (the normal has a positive **k** component). Find the flux of $\mathbf{F} = \langle x,\ y,\ z \rangle$ through S.

Solution P5

Flux through a graph, upward normal

Orientation: upward. The surface is the graph of $g = 4 - x^2 - y^2$, with $g_x = -2x$, $g_y = -2y$, over the disk $R : x^2 + y^2 \le 4$ (where $z \ge 0$). The upward normal satisfies

$$\mathbf{n}\, dS = \langle -g_x,\ -g_y,\ 1\rangle\, dA = \langle 2x,\ 2y,\ 1\rangle\, dA$$

Its $\mathbf{k}$ component is $+1$, so this is the upward side. Put $z = 4 - x^2 - y^2$ into $\mathbf{F}$:

$$\mathbf{F} \cdot \mathbf{n}\, dS = \left(2x^2 + 2y^2 + 4 - x^2 - y^2\right) dA = \left(x^2 + y^2 + 4\right) dA$$

$$\text{Flux} = \int_0^{2\pi}\int_0^2 (r^2 + 4)\, r\, dr\, d\theta = 2\pi\Big[\frac{r^4}{4} + 2r^2\Big]_0^2 = 2\pi(4 + 8)$$

$$\boxed{\text{Flux} = 24\pi \approx 75.40 \quad \text{(upward)}}$$

Positive flux is what you expect: $\mathbf{F}$ points away from the origin, and the upward side of the cap faces away from the origin. Reversing the orientation to downward would give -24π.

Problem P6

Let S be the sphere $x^2 + y^2 + z^2 = 4$, oriented **outward**. Find the flux of $\mathbf{F} = \langle x,\ y,\ z^2 \rangle$ through S.

Solution P6

Flux through a closed surface, parametric

Orientation: outward, away from the origin. Parametrize with the spherical angles (φ measured from the positive z-axis):

$$\mathbf{r}(\varphi,\theta) = \langle 2\sin\varphi\cos\theta,\ 2\sin\varphi\sin\theta,\ 2\cos\varphi\rangle, \qquad 0 \le \varphi \le \pi,\ \ 0 \le \theta \le 2\pi$$

$$\mathbf{r}_\varphi = \langle 2\cos\varphi\cos\theta,\ 2\cos\varphi\sin\theta,\ -2\sin\varphi\rangle, \qquad \mathbf{r}_\theta = \langle -2\sin\varphi\sin\theta,\ 2\sin\varphi\cos\theta,\ 0\rangle$$

$$\mathbf{r}_\varphi \times \mathbf{r}_\theta = 4\sin\varphi\,\langle \sin\varphi\cos\theta,\ \sin\varphi\sin\theta,\ \cos\varphi\rangle$$

Direction check: this is a positive multiple of $\mathbf{r}$ itself for $0 < \varphi < \pi$, so it points outward and no sign flip is needed. On S, $\mathbf{F} = \langle 2\sin\varphi\cos\theta,\ 2\sin\varphi\sin\theta,\ 4\cos^2\varphi\rangle$, so

$$\mathbf{F}\cdot(\mathbf{r}_\varphi \times \mathbf{r}_\theta) = 4\sin\varphi\Big[2\sin^2\varphi\cos^2\theta + 2\sin^2\varphi\sin^2\theta + 4\cos^3\varphi\Big] = 8\sin^3\varphi + 16\sin\varphi\cos^3\varphi$$

$$\text{Flux} = \int_0^{2\pi}\!\!\int_0^{\pi} \left(8\sin^3\varphi + 16\sin\varphi\cos^3\varphi\right) d\varphi\, d\theta$$

With $\int_0^\pi \sin^3\varphi\, d\varphi = \frac{4}{3}$ and $\int_0^\pi \sin\varphi\cos^3\varphi\, d\varphi = \Big[-\frac{\cos^4\varphi}{4}\Big]_0^\pi = 0$:

$$\text{Flux} = 2\pi\Big(8\cdot\frac{4}{3} + 0\Big)$$

$$\boxed{\text{Flux} = \frac{64\pi}{3} \approx 67.02 \quad \text{(outward)}}$$

The z^2 component contributes nothing: $F_z n_z = z^2\cdot z/2$ is odd in z, so the upper and lower halves of the sphere cancel. All of the flux comes from the $\langle x, y\rangle$ part.

Problem P7

Let $\mathbf{F} = \langle y^2 + z,\ x,\ z^2 \rangle$ and let S be the part of the paraboloid $z = 4 - x^2 - y^2$ with $z \geq 0$, oriented upward. Let C be its boundary. Verify Stokes' theorem by computing both $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS$ and $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

Solution P7

Both sides of Stokes' theorem

Hypotheses. S is a smooth surface (a graph of a polynomial), oriented upward. Its boundary is the circle C: $x^2 + y^2 = 4$, $z = 0$, a simple closed smooth curve. $\mathbf{F}$ is polynomial, so it has continuous first partials on all of $\mathbb{R}^3$. Stokes' theorem applies.

Orientation of C. Upward $\mathbf{n}$ and the right-hand rule give counterclockwise travel on C seen from above: $\mathbf{r}(t) = \langle 2\cos t,\ 2\sin t,\ 0\rangle$, $0 \le t \le 2\pi$.

Surface side. With $P = y^2 + z$, $Q = x$, $R = z^2$:

$$\nabla \times \mathbf{F} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle = \langle 0,\ 1,\ 1 - 2y\rangle$$

For the graph $g = 4 - x^2 - y^2$, the upward $\mathbf{n}\,dS = \langle 2x,\ 2y,\ 1\rangle\,dA$, so

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \big(0 + 2y + 1 - 2y\big)\,dA = dA$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \iint_{x^2+y^2\le 4} 1\,dA = 4\pi$$

Line side. On C, $z = 0$ and $dz = 0$, so $\mathbf{F} \cdot d\mathbf{r} = y^2\,dx + x\,dy$. With $x = 2\cos t$, $y = 2\sin t$, $dx = -2\sin t\,dt$, $dy = 2\cos t\,dt$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \Big[-8\sin^3 t + 4\cos^2 t \Big]\,dt = 0 + 4\pi$$

since $\int_0^{2\pi} \sin^3 t\,dt = 0$ and $\int_0^{2\pi} \cos^2 t\,dt = \pi$.

$$\boxed{\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = 4\pi = \oint_C \mathbf{F} \cdot d\mathbf{r}}$$

Both sides agree. Had C been traversed clockwise, the line integral would have come out as -4π, which is how a wrong orientation shows up.

Problem P8

Let $\mathbf{F} = \langle -yz,\ xz,\ xy \rangle$ and let S be the cap of the sphere $x^2 + y^2 + z^2 = 25$ with $z \geq 3$, oriented outward (away from the origin). Find $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS$, using Stokes' theorem to replace S by a flat surface.

Solution P8

Swap the curved surface for a disk

Boundary. The sphere meets the plane $z = 3$ where $x^2 + y^2 = 16$. So C is the circle of radius 4 centered on the z-axis at height $z = 3$. The outward normal on the cap has a positive $\mathbf{k}$ component, so by the right-hand rule C runs counterclockwise seen from above.

Hypotheses. $\mathbf{F}$ is polynomial, so it is C^1 on all of $\mathbb{R}^3$. The cap is smooth and C is a simple closed smooth curve. Stokes' theorem applies to S, and equally to the flat disk D in the plane $z = 3$ bounded by the same circle C, oriented upward so that C keeps the same direction. Since $\nabla \times \mathbf{F}$ is continuous on the solid region between S and D as well, the two surface integrals are equal to the same line integral.

Curl. With $P = -yz$, $Q = xz$, $R = xy$:

$$\nabla \times \mathbf{F} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y \rangle = \langle x - x,\ -y - y,\ z + z \rangle = \langle 0,\ -2y,\ 2z \rangle$$

Swap to the disk. On D, $\mathbf{n} = \mathbf{k}$ and $dS = dA$, so only the $\mathbf{k}$ component of the curl matters, and $z = 3$ there:

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = 2z = 6$$

$$\iint_D 6\, dA = 6 \cdot \pi \cdot 4^2 = 96\pi$$

$$\boxed{\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dA = 96\pi \approx 301.6}$$

Check on the line integral. On C, $\mathbf{r} = \langle 4\cos t,\ 4\sin t,\ 3 \rangle$, and $\mathbf{F} \cdot d\mathbf{r} = z\,(x\,dy - y\,dx) = 3 \cdot 16\, dt$, so $\oint_C \mathbf{F} \cdot d\mathbf{r} = 48 \cdot 2\pi = 96\pi$.

Doing it directly on the cap is possible, with $\mathbf{n} = \frac{1}{5}\langle x, y, z \rangle$ and $(\nabla \times \mathbf{F}) \cdot \mathbf{n} = \frac{2}{5}(z^2 - y^2)$ integrated over spherical angles, but it takes far longer and returns the same 96π. The flat disk replaces a trigonometric integral over a curved cap with one line of arithmetic.